MICROECONOMICS, Spring 2026

Problem Set 1

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1 Utility functions

Do the following utility functions represent the same preferences? Please provide your reasoning.

(a) $u(x, y) = xy, \quad v(x, y) = 3(xy)^2 + 6$
(b) $u(x, y) = xy, \quad v(x, y) = -3(xy)^2 + 6$
(c) $u(x, y) = xy, \quad v(x, y) = \ln x + \ln y$
(d) $u(x, y) = xy, \quad v(x, y) = x + y$

Answer: Here is the answer.

2 Indifference curves

Well-behaved indifference curves (monotonic and convex) are downward sloping. True or false? Please prove your answer.

Answer: Here is the answer.

3 Budget Sets

This question concerns a consumer who is choosing how many of two goods to buy: footballs (the round ones, that you kick with your foot) and cricket balls (like baseballs, but better). The consumer has an income of \$20, and the cost of a football is \$4 and a cricket ball is \$2.

1. Write down the equation for the consumer's budget constraint and graph it in the commodity space.

2. The government decides that football is evil and needs to be taxed. They introduce a 50% tax on each football sold. Rewrite and re-graph the budget constraint.
3. A new government is elected that hates all sports. They now tax both footballs and cricket balls at 50%. What does the budget constraint look like now?
4. Due to a threat of revolt amongst sports fans, the government hands out a subsidy of \$10 to the consumer. What does their new budget constraint look like? How would you expect consumer behavior to differ between this situation and the no-tax, no-subsidy situation described in part 1?
5. Revolution comes, and all taxes and subsidies are abolished. Even better, the consumer finds a new shop that offers bulk discounts. In this shop, footballs cost \$4 each if you buy 3 or fewer. However, the cost of any additional football after 3 is \$2. What does the budget set look like now? Graph it in the commodity space.

Answer: Here is the answer.

4 UMP

Edmund consumes two commodities, sewage and punk rock video cassettes. He does not drink sewage, of course, but he gets paid for taking it away at \$2 per ton. Edmund can accept as much sewage as he wishes at that price. He has no other source of income. Video cassettes cost him \$6 each. He has a utility function $u(s, v)$ on sewage (s) and videos (v) which is decreasing in s and increasing in v.

1. If Edmund accepts 0 tons of sewage, how many video cassettes can he buy?
2. Write down Edmund's constrained optimization problem.
3. Draw Edmund's budget line and shade his budget set.

Answer: Here is the answer.

5 UMP

A consumer has utility function $u(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}}$. Let the prices of goods x and y be given by p_x and p_y respectively, and let m denote the consumer's wealth. Assume that p_x, p_y, and m are all strictly positive.

1. Write down the consumer's utility maximization problem.
2. Derive the consumer's optimal choice of x and y.
3. Are there any prices and wealth (p_x, p_y, m) at which good x is a Giffen good? Briefly explain why.

Answer: Here is the answer.